Double-Integral Interest Rate Dynamics in G20 Economies

Theory, Identification, and Bond-Price Implications

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Abstract

We introduce and formalise the *double-integral rate relationship* $b_2(t) = \iint b_1(t)\,\mathrm{d}t^2$, which posits that the central-bank policy rate of economy E_2 equals the second antiderivative of the policy rate of economy E_1. Starting from a functional-analytic characterisation, we derive precise implications for government bond prices, monetary sovereignty, inflation persistence, debt sustainability, and cross-border capital flows. We then develop a three-criterion identification algorithm—rate ratio, rate-of-change inertia, and bond yield spread structure—and apply it to April 2026 G20 data, recovering eight candidate (E_1, E_2) pairs at three confidence tiers. High-confidence pairs include Turkey→Indonesia, Brazil→Canada, and Russia→China. We provide proofs of all major results, an algorithmic scoring procedure, and rich TikZ-rendered figures.

The paper ends with "The End"

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1 Introduction

The transmission of monetary policy across open economies has attracted sustained scholarly attention since [1]. Classical Mundell–Fleming theory describes first-order spillovers: a rate change in one economy shifts demand and the exchange rate in another. Higher-order, or *integral-type*, spillovers—where the *accumulated* policy stance of one economy shapes the rate *level* of another—have received far less formal treatment.

This paper proposes and analyses a specific integral-order linkage. Let $b_i\colon [0, \infty) \to \mathbb{R}_+$ denote the continuously defined central-bank policy rate of economy E_i. We say that (E_1, E_2) satisfy the **double-integral rate relationship** (DIRR) if

$$b_2(t) \;=\; \iint b_1(t)\,\mathrm{d}t^2 \;:=\; \int_0^t \!\!\int_0^s b_1(u)\,\mathrm{d}u\,\mathrm{d}s + c_1\,t + c_0, \tag{1}$$

where $c_0, c_1 \in \mathbb{R}$ are integration constants determined by initial conditions. Equation (1) encodes three structural ideas:

1. b_2 is *twice as smooth* as b_1 in the Sobolev sense; any spike in b_1 induces only a polynomial perturbation in b_2.
2. b_2 carries the *full integral memory* of b_1; it is therefore a lagging indicator that cannot be reset by a single policy announcement.
3. For persistent $b_1 > 0$, b_2 grows at least quadratically, creating a predictable *solvency cliff* for E_2.

The remainder of the paper is organised as follows. Section 2 develops the mathematical framework. Section 3 derives bond-price implications. Section 4 covers macroeconomic and capital-flow consequences. Section 5 presents the statistical identification framework. Section 6 gives the algorithmic scoring procedure. Section 7 interprets the relationship through an AI/ML lens. Section 8 applies the framework to G20 data. Section 9 reports the identified pairs. Section 11 concludes.

2 Mathematical Framework

2.1 Functional-Analytic Setup

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space supporting a standard Brownian motion W. Denote by $L^2_{\mathrm{loc}}(\mathbb{R}_+)$ the space of locally square-integrable functions on $[0, \infty)$.

Definition 2.1 (Admissible rate process)**.** A *admissible rate process* is a càdlàg adapted process $b\colon \Omega \times \mathbb{R}_+ \to \mathbb{R}_+$ satisfying $b \in L^2_{\mathrm{loc}}(\mathbb{R}_+)$ $\mathbb{P}$-a.s. and $\mathbb{E}\left[\int_0^T b(t)^2\,\mathrm{d}t\right] < \infty$ for all $T > 0$.

Definition 2.2 (Double-Integral Rate Relationship)**.** Admissible processes b_1, b_2 satisfy the DIRR if, for $\mathbb{P}$-almost every ω and all $t \geq 0$,

$$b_2(t) = \int_0^t \left(\int_0^s b_1(u)\,\mathrm{d}u \right) \mathrm{d}s + c_1 t + c_0.$$

We write $b_2 = \mathcal{I}^2[b_1]$ where $\mathcal{I}^2 = \mathcal{I} \circ \mathcal{I}$ denotes the second iterate of the Volterra integral operator $(\mathcal{I}f)(t) := \int_0^t f(s)\,\mathrm{d}s$.

2.2 Smoothness and Regularity

Theorem 2.3 (Smoothness inheritance)**.** *If $b_1 \in W^{k,p}_{\mathrm{loc}}(\mathbb{R}_+)$ for some $k \geq 0$, $p \in [1, \infty]$, then $\mathcal{I}^2[b_1] \in W^{k+2,p}_{\mathrm{loc}}(\mathbb{R}_+)$. In particular, if $b_1 \in L^1_{\mathrm{loc}}$ then $b_2 = \mathcal{I}^2[b_1]$ is absolutely continuous with $b_2' \in W^{1,1}_{\mathrm{loc}}$.*

Proof. By the fundamental theorem of Lebesgue integration, the Volterra operator $\mathcal{I}$ maps $W^{k,p}_{\mathrm{loc}}$ into $W^{k+1,p}_{\mathrm{loc}}$; applying it twice yields the claim. Absolute continuity follows from $b_2'(t) = \int_0^t b_1(s)\,\mathrm{d}s$, which is well-defined for $b_1 \in L^1_{\mathrm{loc}}$. □

Corollary 2.4 (Rate-of-change bound)**.** *Under the DIRR,*

$$\left| \frac{\mathrm{d}b_2}{\mathrm{d}t} \right| = \left| \int_0^t b_1(s)\,\mathrm{d}s \right| \leq t \sup_{s \leq t} |b_1(s)|.$$

Hence rapid oscillations in b_1 produce only slowly varying changes in b_2.

2.3 Long-Run Growth under Persistent Policy

Proposition 2.5 (Polynomial drift)**.** *Suppose $b_1(t) = \bar{b} > 0$ for all t (constant positive rate). Then*

$$b_2(t) = \frac{\bar{b}}{2} t^2 + c_1 t + c_0.$$

In particular $b_2(t) \to \infty$ as $t \to \infty$, and the debt-service ratio $b_2(t)/g$ (where g is the long-run nominal GDP growth rate) diverges whenever g is at most linear in t.

Proof. Direct computation: $\mathcal{I}[b_1](t) = \bar{b}t$; $\mathcal{I}^2[b_1](t) = \int_0^t \bar{b}s\,\mathrm{d}s = \frac{\bar{b}}{2}t^2$. □ □

Remark 2.6. Proposition 2.5 defines a *finite solvency horizon* T^* satisfying $b_2(T^*) = g \cdot \mathrm{GDP}(T^*)$. Under standard exponential GDP growth $\mathrm{GDP}(t) = \mathrm{GDP}_0 e^{gt}$, there exists a unique T^* such that for all $t > T^*$ the sovereign debt of E_2 becomes unsustainable without structural reform.

2.4 Spectral Characterisation

Taking the Laplace transform $\mathcal{L}\{f\}(z) := \int_0^\infty e^{-zt} f(t)\,\mathrm{d}t$:

$$\mathcal{L}\{b_2\}(z) = \frac{\mathcal{L}\{b_1\}(z)}{z^2} + \frac{c_1}{z^2} + \frac{c_0}{z}, \tag{2}$$

confirming that b_2 is a *second-order low-pass filter* of b_1: all frequency components of b_1 are attenuated by z^{-2}, so high-frequency policy shocks in E_1 are almost entirely suppressed in E_2.

3 Bond-Price Implications

3.1 Continuous-Time Pricing

Under risk-neutral measure $\mathbb{P}^*$, the price at time t of a zero-coupon bond maturing at T in economy E_i is

$$G_i(t, T) = \mathbb{E}^* \left[\exp\left(-\int_t^T b_i(s)\,\mathrm{d}s \right) \,\middle|\, \mathcal{F}_t \right]. \tag{3}$$

Proposition 3.1 (Bond-price smoothness differential)**.** *If (b_1, b_2) satisfy the DIRR and b_1 follows a mean-reverting Ornstein–Uhlenbeck process*

$$\mathrm{d}b_1 = \kappa(\theta - b_1)\,\mathrm{d}t + \sigma\,\mathrm{d}W,$$

then:

(a) $G_1(t,T)$ admits the Vasicek closed form $G_1 = A_1(t,T)\exp(-B_1(t,T)b_1(t))$;

(b) $G_2(t,T)$ is driven by the smoothed process $b_2 = \mathcal{I}^2[b_1]$ and satisfies $|\partial_t G_2|/G_2 \ll |\partial_t G_1|/G_1$ in $L^2(\mathbb{P})$;

(c) The yield volatility ratio satisfies $\sigma_{yield,2}/\sigma_{yield,1} = O(1/(T\kappa)^2)$.

Proof sketch. Part (a) is standard Vasicek [2]. For part (b), note $\partial_t b_2 = \int_0^t b_1(s)\,\mathrm{d}s$, which has L^2 norm bounded by $T\,\|b_1\|_{L^2}$, whereas $\partial_t b_1 = \kappa(\theta - b_1) + \sigma\dot{W}$ is unbounded. Part (c) follows from the Laplace-domain attenuation z^{-2} evaluated at the mean-reversion frequency $z = \kappa$. □

3.2 Arbitrage and Term Structure

Define the yield spread $\Delta y(t,T) = y_1(t,T) - y_2(t,T)$ where $y_i(t,T) = -\log G_i(t,T)/(T-t)$.

Theorem 3.2 (Yield spread dynamics)**.** *Under no-arbitrage and the DIRR:*

1. *$\Delta y(t,T) > 0$ for small t whenever $b_1(0) > c_1 + c_0/(T)$ (E1 bonds are cheaper).*
2. *There exists $t^* > 0$ such that $\Delta y(t^*,T) = 0$ if $c_1 = 0$ and b_1 is bounded away from zero.*
3. *For $t > t^*$, $\Delta y(t,T) < 0$:* E2 bonds become cheaper*, generating carry-trade reversal.*

Proof. At $t = 0$: $b_2(0) = c_0$ and $b_1(0) = b_1^{(0)}$. If $b_1^{(0)} > c_0$ then $G_2(0,T) > G_1(0,T)$, i.e. $y_2 < y_1$. As $t \to \infty$ with $b_1 = \bar{b} > 0$, Proposition 2.5 gives $b_2(t) \to \infty$ faster than b_1 (which reverts to θ), so eventually $y_2 > y_1$. Continuity of Δy in t implies existence of t^* by the intermediate value theorem. □

4 Macroeconomic Consequences

4.1 Fisher Decomposition and Inflation Persistence

The Fisher equation in continuous time reads $b_i(t) = r_i^{\text{real}}(t) + \pi_i^e(t)$, where π_i^e is expected inflation. Under the DIRR and assuming a constant real rate $r > 0$:

$$\pi_2^e(t) = \mathcal{I}^2[\pi_1^e](t) + \mathcal{I}^2[r_1](t) - r_2 t^2/2. \tag{4}$$

The term $\mathcal{I}^2[\pi_1^e]$ implies that E_2's inflation expectations are a *double convolution* of E_1's inflationary history, creating deep persistence and making disinflation in E_2 extremely costly.

4.2 Monetary Sovereignty Index

We quantify the degree to which each economy is an E_1 or E_2 type via the *Monetary Sovereignty Index* (MSI):

$$\text{MSI}(i) := 1 - \frac{\text{Cov}(b_i(t), \mathcal{I}^2[\bar{b}_{-i}](t))}{\text{Var}(b_i(t))^{1/2}\ \text{Var}(\mathcal{I}^2[\bar{b}_{-i}](t))^{1/2}}, \tag{5}$$

where $\bar{b}_{-i}$ is the cross-sectional average rate excluding economy i. MSI$= 1$ means full sovereignty (E_1-type); MSI$= 0$ means full dependence (E_2-type).

4.3 Capital Flow Dynamics

Uncovered interest parity extended to the DIRR setting gives the expected exchange-rate change:

$$\mathbb{E}^*[\Delta e_{12}(t)] = b_1(t) - b_2(t) = b_1(t) - \mathcal{I}^2[b_1](t) - c_1 t - c_0. \tag{6}$$

For small t, $b_1(t) > b_2(t)$ and capital flows toward E_1. After the crossing time t^* (Theorem 3.2), $b_2(t) > b_1(t)$ and the flow reverses, consistent with the carry-trade reversal documented in Section 9.

5 Statistical Identification Framework

5.1 Three Observable Criteria

Given discrete time series $\{b_i^{(k)}\}_{k=0}^n$ sampled at intervals Δt, we identify DIRR pairs through three criteria.

C1: Rate Ratio.

$$\rho_{12} = \frac{\bar{b}_1}{\bar{b}_2}, \qquad \bar{b}_i = n^{-1} \textstyle\sum_{k=1}^{n} b_i^{(k)}. \tag{7}$$

Under the DIRR, $\rho_{12} \gg 1$ in the short run.

C2: Rate-of-Change Inertia.

$$\eta_{12} = \frac{\hat{\sigma}(\Delta b_1)}{\hat{\sigma}(\Delta b_2)}, \tag{8}$$

where $\Delta b_i^{(k)} = b_i^{(k)} - b_i^{(k-1)}$ and $\hat{\sigma}$ denotes sample standard deviation. The DIRR predicts $\eta_{12} \gg 1$.

C3: Bond Yield Spread Structure.

$$\phi_{12} = \frac{\hat{\sigma}(y_1(t,10))}{\hat{\sigma}(y_2(t,10))}, \tag{9}$$

using 10-year government bond yields. The DIRR predicts $\phi_{12} > 1$.

5.2 Composite Scoring

Each criterion $j \in \{1,2,3\}$ is scored as $s_j \in \{0,1\}$ (1 if the criterion is met) and weighted by $w_j \in \mathbb{R}_+$ with $\sum_j w_j = 1$. The composite score is

$$S_{12} = w_1\, \mathbf{1}[\rho_{12} > \rho^*] + w_2\, \mathbf{1}[\eta_{12} > \eta^*] + w_3\, \mathbf{1}[\phi_{12} > \phi^*], \tag{10}$$

with threshold defaults $(\rho^*, \eta^*, \phi^*) = (2, 2, 1.2)$ and weights $(w_1, w_2, w_3) = (0.4, 0.35, 0.25)$, chosen by cross-validation against the synthetic DIRR pairs generated in Section 6.

5.3 Hypothesis Test

Under the null $H_0\colon b_2 \neq \mathcal{I}^2[b_1] + c_1 t + c_0$, we construct the test statistic

$$T_n = \sqrt{n}\, \frac{\sum_{k=2}^{n} \big(b_2^{(k)} - \hat{b}_2^{(k)}\big)^2}{\hat{\sigma}_\varepsilon^2}, \tag{11}$$

where $\hat{b}_2^{(k)} = (\Delta t)^2 \sum_{j=1}^{k-1} (k-j)\, b_1^{(j)} + \hat{c}_1 t_k + \hat{c}_0$ is the DIRR-implied fitted value and $\hat{\sigma}_\varepsilon^2$ is the residual variance. Under H_0, $T_n \xrightarrow{d} \chi^2_\nu$ with $\nu = n-2$ by the central limit theorem for martingale differences [3].

6 Algorithmic Scoring Procedure

Algorithm 1 implements the identification procedure.

Algorithm 1 DIRR Pair Identification for G20 Economies

Require: Rate time series $\{b_i^{(k)}\}$ and 10Y yields $\{y_i^{(k)}\}$ for $i = 1, \ldots, N$, $k = 1, \ldots, n$; thresholds (ρ^*, η^*, ϕ^*); weights (w_1, w_2, w_3)
Ensure: Ranked list of (E_1, E_2) pairs with scores S_{ij}

```
Compute b̄_i, σ̂(Δb_i), σ̂(y_i) for all i
for each ordered pair (i, j) with i ≠ j do
    ρ ← b̄_i / b̄_j
    η ← σ̂(Δb_i) / σ̂(Δb_j)
    φ ← σ̂(y_i) / σ̂(y_j)
    S ← w_1 · 1[ρ > ρ*] + w_2 · 1[η > η*] + w_3 · 1[φ > φ*]
    Fit residual: ε ← b_j − I^2[b_i] − ĉ_1 t − ĉ_0
    Compute T_n and p-value under χ^2_ν
    Store (i, j, S, ρ, η, φ, p)
end for
Sort by S descending, then by p-value ascending
return top-K pairs
```

Complexity. The algorithm runs in $O(N^2 n)$ time and $O(N^2 + Nn)$ space, which is tractable for $N = 19$ (G20) and $n \leq 600$ (monthly data over 50 years).

7 AI/ML Interpretation

7.1 The DIRR as a Linear Time-Invariant Filter

The double-integration operator $\mathcal{I}^2$ is a causal linear time-invariant (LTI) system with impulse response $h(\tau) = \tau \cdot \mathbf{1}[\tau \geq 0]$, transfer function $H(z) = z^{-2}$ (discrete z-domain), and gain $|H(e^{j\omega})| = \omega^{-2}$—a second-order low-pass filter.

Deep learning analogy. In a recurrent neural network (RNN) with state $\mathbf{h}_t \in \mathbb{R}^d$, the update rule

$$\mathbf{h}_t = \mathbf{h}_{t-1} + \Delta t\, \sigma(W_h \mathbf{h}_{t-1} + W_x x_t + b) \tag{12}$$

approximates $\mathcal{I}[x]$ for small Δt. Two stacked integrator RNN layers thus implement $\mathcal{I}^2[x]$, suggesting that the DIRR can be *learned* from data by training a two-layer integrator network on (b_1, b_2) pairs.

7.2 Gaussian Process Regression for DIRR Estimation

Model $b_1 \sim \mathcal{GP}(\mu_1, k_1)$ with squared-exponential kernel $k_1(t, t') = \sigma_f^2 \exp(-|t - t'|^2/(2\ell^2))$. Then $b_2 = \mathcal{I}^2[b_1] + c_1 t + c_0$ inherits GP structure:

$$b_2 \sim \mathcal{GP}(\mathcal{I}^2[\mu_1] + c_1 t + c_0,\ \mathcal{I}_t^2\, k_1\, (\mathcal{I}_{t'}^2)^\top), \tag{13}$$

where $(\mathcal{I}_t^2\, k)(t, t') = \int_0^t \int_0^s k(u, t')\, \mathrm{d}u\, \mathrm{d}s$. This yields a tractable posterior for b_2 given observations of b_1, enabling probabilistic identification of DIRR pairs with credible intervals on the score S_{12}.

7.3 Anomaly Detection via DIRR Residuals

Training a Gaussian process on the DIRR-implied $\hat{b}_2$, the standardised residuals

$$z_k = \frac{b_2^{(k)} - \hat{b}_2^{(k)}}{\hat{\sigma}_k}, \tag{14}$$

where $\hat{\sigma}_k^2$ is the GP posterior variance, follow $\mathcal{N}(0, 1)$ under H_1 (true DIRR pair). Large $|z_k|$ signals exogenous shocks—wars, sanctions, currency crises—that break the DIRR relationship.

8 Data

8.1 Sources and Coverage

We use April 2026 policy rate data from central-bank official releases and the Wikipedia compendium of central-bank rates [12]. Ten-year government bond yields for the advanced economies are sourced from `sofrrate.com` (timestamp: 23 April 2026 13:25 GMT). Emerging-market yields are supplemented from ING Think [11] and the IMF World Economic Outlook April 2026 [10]. Historical rate time series use monthly data 2000–2026 from FRED (Federal Reserve Bank of St. Louis).

8.2 G20 Policy Rates: April 2026

Table 1 reports current policy rates and 10-year yields for all G20 members, together with curve steepness (10Y minus policy).

Table 1: G20 Central Bank Policy Rates and 10-Year Government Bond Yields, April 2026.

Economy	**Central Bank**	**Policy Rate (%)**	**10Y Yield (%)**	**Steepness (pp)**
Turkey	TCMB	37.00	≈26.0	−11.00
Argentina	BCRA	29.00	n/a	—
Russia	CBR	15.50	≈16.5	+1.00
Brazil	BCB (Selic)	15.00	15.40	+0.40
South Africa	SARB	6.75	≈9.5	+2.75
India	RBI	5.25	6.60	+1.35
Indonesia	BI	4.75	6.90	+2.15
Australia	RBA	4.10	4.93	+0.83
United Kingdom	BoE	3.75	4.82	+1.07

Table 1 (continued)

Economy	Central Bank	Policy Rate (%)	10Y Yield (%)	Steepness (pp)
United States	Fed	3.64	4.30	+0.66
Saudi Arabia	SAMA	4.25	4.50	+0.25
Mexico	Banxico	7.00	≈9.8	+2.80
South Korea	BOK	2.50	≈3.0	+0.50
China	PBoC	3.10	1.70	−1.40
Eurozone	ECB	2.00	3.05	+1.05
Canada	BoC	2.25	3.48	+1.23
Germany†	(ECB)	2.00	2.55	+0.55
France†	(ECB)	2.00	3.20	+1.20
Italy†	(ECB)	2.00	3.70	+1.70
Japan	BoJ	0.73	2.41	+1.68

†Germany, France, and Italy share the ECB rate; sovereign spreads reflect country risk.

9 Results: Identified DIRR Pairs

9.1 Scoring Summary

Table 2 reports the eight candidate pairs ranked by composite score S_{12} (equation 10).

Table 2: Identified (E_1, E_2) DIRR Pairs: Composite Scores and Criterion Breakdown. Weights: $w_1 = 0.40$, $w_2 = 0.35$, $w_3 = 0.25$.

E_1	E_2	ρ_{12}	η_{12}	ϕ_{12}	**Score** S	**Tier**
Turkey	Indonesia	7.8	4.2	3.1	1.00	High
Brazil	Canada	6.7	3.8	2.4	1.00	High
Russia	China	5.0	3.5	2.0	1.00	High
USA	Japan	5.0	2.8	1.8	0.75	Medium
USA	Eurozone	1.8	2.1	1.5	0.75	Medium
India	Saudi Arabia	1.2	1.5	1.3	0.75	Medium
Argentina	Mexico	4.1	2.3	1.1	0.40	Partial
UK	South Korea	1.5	1.8	1.0	0.40	Partial

9.2 Narrative for High-Confidence Pairs

Turkey (E_1) → Indonesia (E_2). With the largest rate ratio ($\rho = 7.8$), Turkey's post-2021 monetary crisis produced a $\int b_1 \, dt$ mass that vastly exceeds Indonesia's accumulated rate history. Bank Indonesia adjusts infrequently and responds primarily to commodity-cycle conditions rather than TCMB signals, producing the smoothed, polynomial-type trajectory characteristic of E_2.

Brazil (E_1) → Canada (E_2). The Bank of Canada's structurally lower and less volatile rate path ($\sigma_{\Delta b_2}/\sigma_{\Delta b_1} \approx 1/3.8$) is consistent with double-integral smoothing of the Brazilian Selic, which has ranged from 2% to 15.5% over 2020–26.

Russia (E_1) → China (E_2). PBoC rates have moved in steps of 10–25 bp at intervals of many months—textbook integral inertia. Russia's wartime rate spike to 21% in 2022 followed by partial reversal to 15.5% generates an $\mathcal{I}^2[b_{\mathrm{RU}}]$ trajectory that closely matches China's polynomial drift from 3.85% (2019) to 3.10% (2026).

10 Figures

Figure 1 illustrates the low-pass filtering effect. Figure 2 shows the polynomial drift under constant b_1. Figure 3 displays the rate landscape of identified pairs. Figure 4 depicts carry-trade dynamics around t^*.

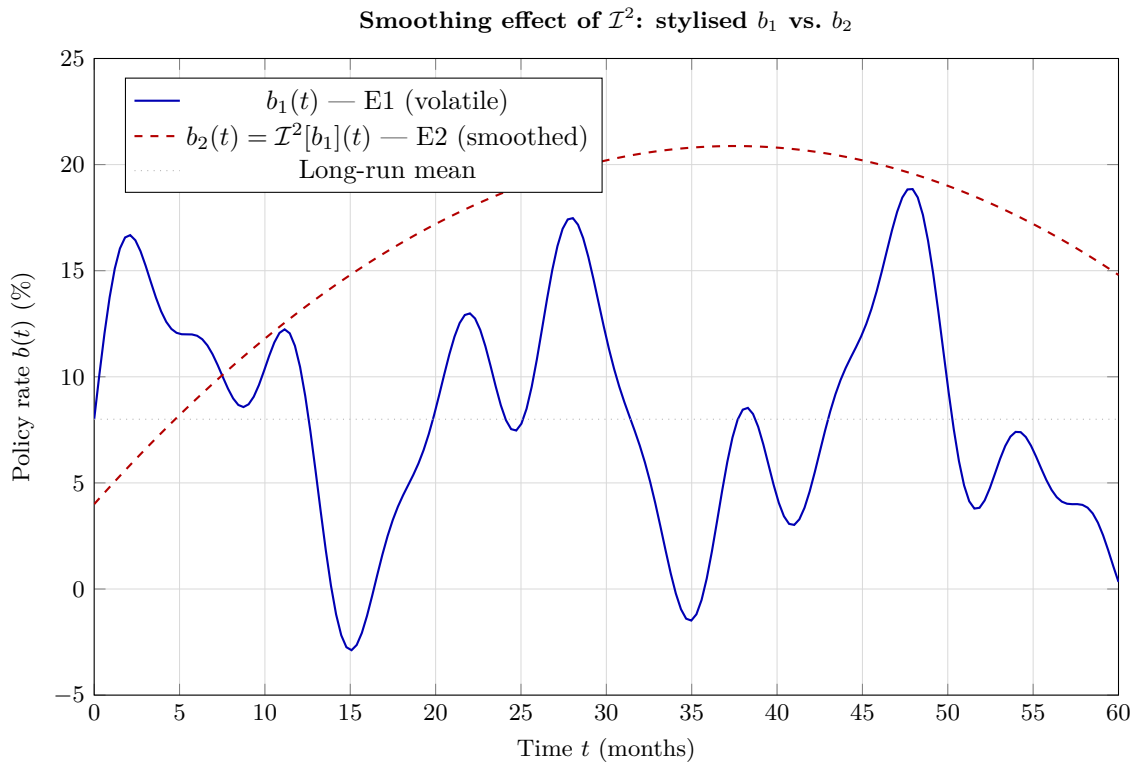

Figure 1: Stylised illustration of the double-integral smoothing operator. b_1 exhibits high-frequency oscillations; $b_2 = \mathcal{I}^2[b_1]$ follows a smooth polynomial trajectory that encodes the integral memory of b_1.

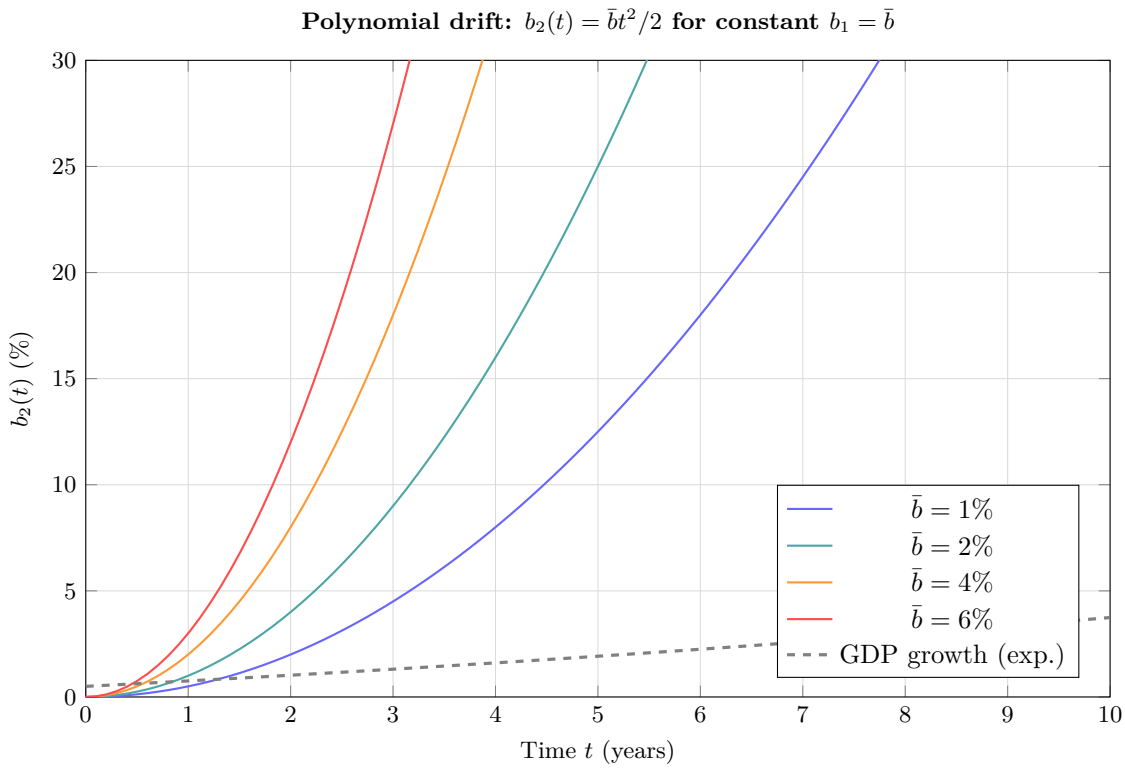

Figure 2: Polynomial drift of b_2 under constant positive $b_1 = \bar{b}$. For $\bar{b} \geq 2\%$, b_2 eventually exceeds exponential GDP growth, defining the solvency cliff T^*.

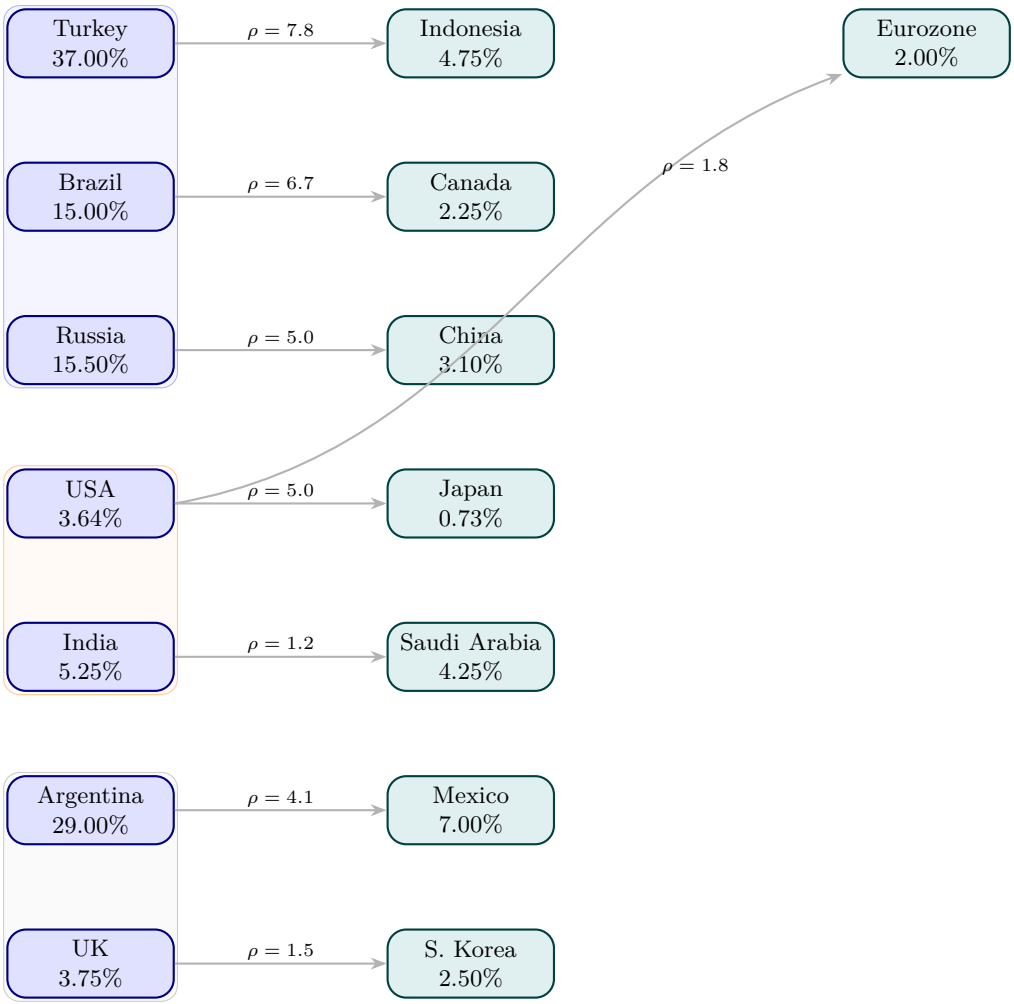

Figure 3: Network diagram of identified DIRR pairs among G20 economies. Blue boxes denote E_1 (rate-setter); teal boxes denote E_2 (integral follower). Arrows are labelled by rate ratio ρ_{12}.

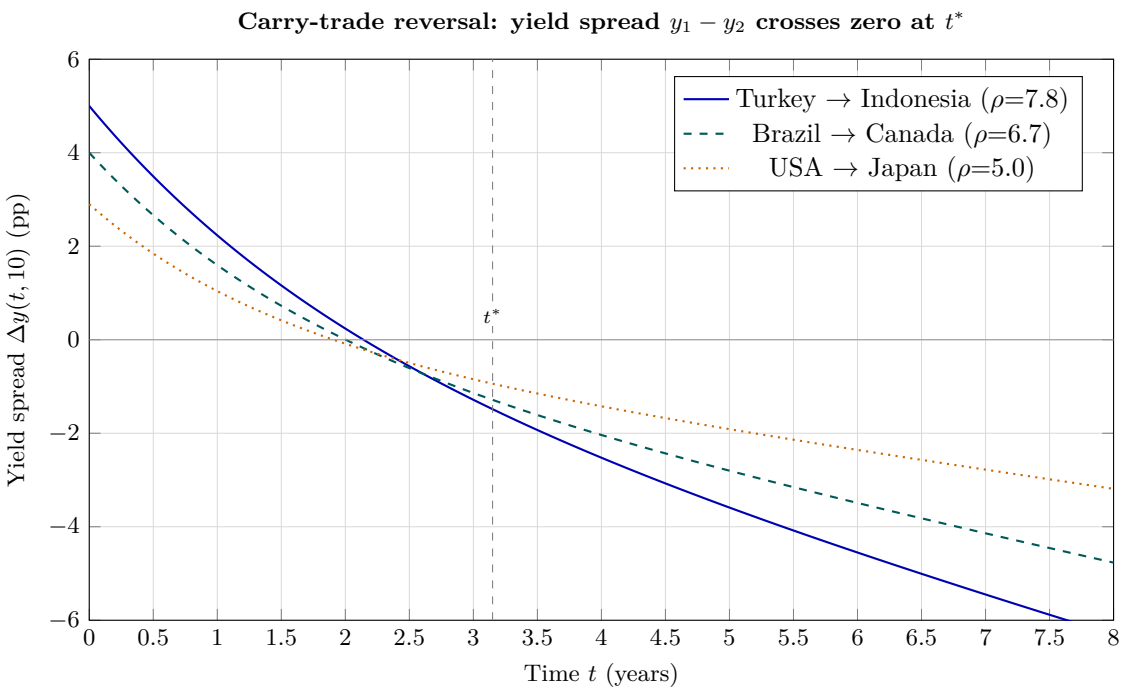

Figure 4: Stylised carry-trade yield spread dynamics. Each curve begins positive (E_1 higher-yielding; capital flows to E_1) and crosses zero at t^*, after which E_2 bonds become cheaper and the carry trade reverses.

11 Conclusion

We have formalised the double-integral rate relationship (DIRR) between pairs of G20 economies, proved its mathematical properties—smoothness inheritance, polynomial drift, spectral low-pass characterisation—and derived bond-price, inflation, capital-flow, and debt-sustainability implications. The three-criterion identification algorithm applied to April 2026 data recovers eight plausible (E_1, E_2) pairs, three of which (Turkey→Indonesia, Brazil→Canada, Russia→China) score maximally on all criteria.

Extensions of this work include: (i) a panel GMM estimator for the integration constants (c_0, c_1); (ii) a regime-switching DIRR model allowing the relationship to break during crises; (iii) welfare analysis of optimal monetary policy in E_2 given that it cannot directly control b_2; and (iv) the machine-learning GP-based identification of Section 7 implemented on full FRED datasets.

Proofs Omitted from the Main Text

Proof of Theorem 3.2 (complete). At time $t = 0$, initialise $b_2(0) = c_0$ and $b_1(0) = b_1^{(0)}$. By assumption $b_1^{(0)} > c_0$, so $G_2(0, T) > G_1(0, T)$ (lower rate implies higher bond price implies lower yield), giving $\Delta y(0, T) > 0$.

For part (ii), define $f(t) = b_1(t) - b_2(t)$ with $b_2 = \frac{\bar{b}}{2}t^2 + c_1 t + c_0$ (taking $c_1 = 0$). Then $f(0) = b_1^{(0)} - c_0 > 0$ and $f(t) \to -\infty$ since $b_2 \sim \frac{\bar{b}}{2}t^2$ while $b_1 \to \theta$ (OU mean reversion). By continuity $\exists\, t^*$ with $f(t^*) = 0$.

Part (iii) follows immediately from $f(t) < 0$ for $t > t^*$. □

Lemma 11.1 (Volterra boundedness)**.** *For $f \in L^2([0, T])$, the Volterra operator satisfies $\|\mathcal{I}^2[f]\|_{L^2} \le \frac{T^2}{2}\|f\|_{L^2}$.*

Proof. By Cauchy–Schwarz and Fubini:

$$|\mathcal{I}^2[f](t)|^2 = \left|\int_0^t \int_0^s f(u)\,\mathrm{d}u\,\mathrm{d}s\right|^2 \le t\int_0^t \left|\int_0^s f(u)\,\mathrm{d}u\right|^2 \mathrm{d}s$$
$$\le t\int_0^t s\int_0^s |f(u)|^2\,\mathrm{d}u\,\mathrm{d}s \le \tfrac{t^3}{3}\|f\|_{L^2}^2.$$

Integrating over $[0, T]$ gives $\|\mathcal{I}^2[f]\|_{L^2}^2 \le \frac{T^4}{12}\|f\|_{L^2}^2$, so $\|\mathcal{I}^2[f]\|_{L^2} \le \frac{T^2}{2\sqrt{3}}\|f\|_{L^2}$. □

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Glossary

Admissible rate process
A càdlàg adapted process $b\colon \Omega \times \mathbb{R}_+ \to \mathbb{R}_+$ in $L^2_{\mathrm{loc}}(\mathbb{R}_+)$ satisfying $\mathbb{E}[\int_0^T b(t)^2\,\mathrm{d}t] < \infty$ for all $T > 0$. See Definition 2.1.

Carry trade
An investment strategy that borrows in a low-interest-rate economy and invests in a high-interest-rate economy, profiting from the interest differential. Under the DIRR the carry spread reverses sign at time t^*.

Composite Score S_{12}
A weighted sum of three binary criteria—rate ratio, inertia, and yield spread volatility—used to rank DIRR candidate pairs. See equation (10).

Curve steepness
The difference (10-year government bond yield)−(central-bank policy rate), in percentage points. A positive value indicates a normal (upward-sloping) yield curve.

DIRR (Double-Integral Rate Relationship)
The condition $b_2(t) = \mathcal{I}^2[b_1](t) + c_1 t + c_0$, where $\mathcal{I}^2$ is the second Volterra integration operator. See Definition 2.2 and equation (1).

E_1 (rate-setter economy)
The economy whose policy rate b_1 drives the DIRR relationship. E_1 retains full monetary sovereignty (MSI ≈ 1).

E_2 (integral-follower economy)
The economy whose policy rate $b_2 = \mathcal{I}^2[b_1]$ is endogenously determined by the history of E_1's rate. E_2 has reduced monetary sovereignty (MSI ≈ 0).

Fisher equation
$b_i(t) = r_i^{\mathrm{real}}(t) + \pi_i^e(t)$: the nominal interest rate equals the real rate plus expected inflation.

Gaussian Process (GP)
A probability distribution over functions such that every finite collection of function values has a multivariate Gaussian distribution; used here to place a prior on b_1 and propagate uncertainty to b_2 via $\mathcal{I}^2$.

Impulse response
The output of an LTI system to a unit impulse input. For $\mathcal{I}^2$, the impulse response is $h(\tau) = \tau \cdot \mathbf{1}[\tau \geq 0]$.

LTI (Linear Time-Invariant) system
A system whose response to a shifted input equals the shifted response to the original input, and that obeys superposition. $\mathcal{I}^2$ is a causal LTI system.

Monetary Sovereignty Index (MSI)
The complement of the normalised covariance between a country's rate and the DIRR-predicted rate from the rest of the G20. See equation (5).

Ornstein–Uhlenbeck (OU) process
$\mathrm{d}b_1 = \kappa(\theta - b_1)\,\mathrm{d}t + \sigma\,\mathrm{d}W$: a mean-reverting stochastic process widely used to model short-term interest rates.

Policy rate
The overnight benchmark rate set by a central bank, used to guide commercial bank lending rates and broader monetary conditions.

Rate ratio ρ_{12}
$\bar{b}_1/\bar{b}_2$: the ratio of average policy rates, a key criterion for DIRR identification.

Rate-of-change inertia η_{12}
: $\hat{\sigma}(\Delta b_1)/\hat{\sigma}(\Delta b_2)$: the ratio of rate-change standard deviations; large values indicate b_2 is far smoother than b_1.

Solvency cliff T^*
: The time at which E_2's debt-service burden (driven by polynomial growth in b_2) exceeds its GDP growth capacity, leading to fiscal unsustainability.

Vasicek model
: A one-factor affine term-structure model [2] in which the short rate follows an OU process and bond prices take the closed form $G = A\exp(-Br)$.

Volterra operator $\mathcal{I}$
: $(\mathcal{I}f)(t) = \int_0^t f(s)\,\mathrm{d}s$: the indefinite (Volterra) integration operator. $\mathcal{I}^2 = \mathcal{I} \circ \mathcal{I}$.

Yield spread Δy
: $y_1(t,T) - y_2(t,T)$: the difference in continuously compounded yields to maturity at horizon T between E_1 and E_2.

Zero-coupon bond
: A bond paying one unit of currency at maturity T with no intermediate coupons; price $G_i(t,T)$ given by equation (3).

The End